

Source image



Scene-graph snippet (input to the generator)

objects: {dog (x2), bed, wall, pillow, outlet, sheet, ...}
attributes: wall.color = white; pillow.color = blue; bed = large
relations: dog on bed; pillow on bed; wall to-the-right-of outlet
L1/L3 alternate positive and negative phrasings by image index; L4 reuses L2's options unchanged (v8 design).

L1 — object presence

"Is there a bed in this image?"

gold: (A) yes

L2 — attribute query

"What color is the wall?"

gold: (C) white
options: {white, +3 distractors}

L3 — spatial relation (negative)

"Is the outlet to the right of the wall?"

gold: (B) no

L4 — compositional

"What color is the wall that is to the right of the outlet?"

gold: (C) white
options inherited from L2

L5 — wordy mirror of L1

*"Please read the question carefully and answer it directly.
primary question The extra wording is only polite framing."*

same options, same gold

L6 — wordy mirror of L2

*"Please read the question carefully and answer it directly.
primary question The extra wording is only polite framing."*

same options, same gold

L7 — wordy mirror of L3

*"Please read the question carefully and answer it directly.
primary question The extra wording is only polite framing."*

same options, same gold

L8 — wordy mirror of L4

*"Please read the question carefully and answer it directly.
primary question The extra wording is only polite framing."*

same options, same gold